Statistics: Preliminary Ph.D. Course

Day 3: The Measure-Theoretic Approach to Probability

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Today's Session

Formalizing what we have seen

- Why **Measure Theory**: Limitations of the Riemann Integral. The Dirichlet Function.
- **Measurable Functions** and **Measures**. Dirac, Counting, Lebesgue, and Probability Measures. Definitions, Theorems, Proofs, and Examples.
- Definition of the **Abstract Integral**. From Indicators to Simple Functions and to Measurable Functions. The **Lebesgue Integral** and the **Expectation**. Examples.

If time permits...

- The **Radon-Nikodym Derivative**. Motivation and Theorem.
- Induced Law and **Distribution**.
- **Change of Variables** Formula. Theorem and Proof.
- **PMFs** and **PDFs** as Radon-Nikodým derivatives.

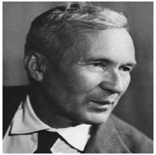
Relation to Measure Theory

- Due to the technicality we mentioned on the first day, we will introduce the approach in **Measure Theory**. This will also enable the unification of both the continuous and discrete cases.
- For that, we need to introduce the **Lebesgue Integral**. It is more general than the **Riemann Integral**, which is the typical one from high school. Hence, we will motivate the new integral based on the limitations of the Riemann integral.
- Keep in mind that an integral is not simply finding an antiderivative. Think about integrals of functions over sets as measures of those functions... More on this now.
- For simplicity, we will focus on the **univariate case**, but everything could be readily extended to higher dimensions.

Motivation

A Bit of History

$$\text{Limitations of Riemann Integral} \underset{Lebesge+Borel}{\Longrightarrow} \text{Measure Theory} \underset{Kolmogorov}{\Longrightarrow} \text{Modern Probability}$$

- **Problem: What is the probability that a number chosen uniformly at random in the unit interval is a rational number?**
- Formally, the question is asking to compute the following probability:

$$\mathbb{P}(X \in \mathbb{Q}) = \int_{\mathbb{Q}} f(x)\ dx = \int_{\mathbb{Q}} \mathbb{1}_{\{0<x<1\}}\ dx = \int_{\mathbb{Q}\cap(0,1)} dx = \int_0^1 \mathbb{1}_{\mathbb{Q}}(x)\ dx$$

Motivation

Limitations of the Riemann Integral

- **Problem: What is the probability that a number chosen uniformly at random in the unit interval is a rational number?**

$$\mathbb{P}(X \in \mathbb{Q}) = \int_0^1 \mathbb{1}_{\mathbb{Q}}(x)\, dx$$

- $\mathbb{1}_{\mathbb{Q}}(x) = \mathbb{1}_{\{x \in \mathbb{Q}\}}$ is the **Dirichlet Function**. It takes value 1 if x is a rational number and 0 otherwise.
- How do we compute this integral? Can we use the Riemann integral?
- Let's go through the basics of the Riemann Integral.

The Riemann Integral

Upper and Lower Sums

- Let f be a non-negative real-valued and bounded function on the interval $[a, b]$, i.e., $f : [a, b] \underset{x \mapsto y}{\longrightarrow} \mathbb{R}$.

- Define a **partition** P of the interval $[a, b]$ as a finite sequence of numbers of the form: $P = \{a = x_0 < x_1 < \cdots < x_i < \cdots < x_n = b\}$. Each $[x_{i-1}, x_i]$ $\forall i \in \{1, \ldots, n\}$ is called a **sub-interval** of the partition.

- For each $i = 1, \ldots, n$, let: $m_i = \inf\limits_{x \in [x_{i-1}, x_i]} f(x)$ and $M_i = \sup\limits_{x \in [x_{i-1}, x_i]} f(x)$

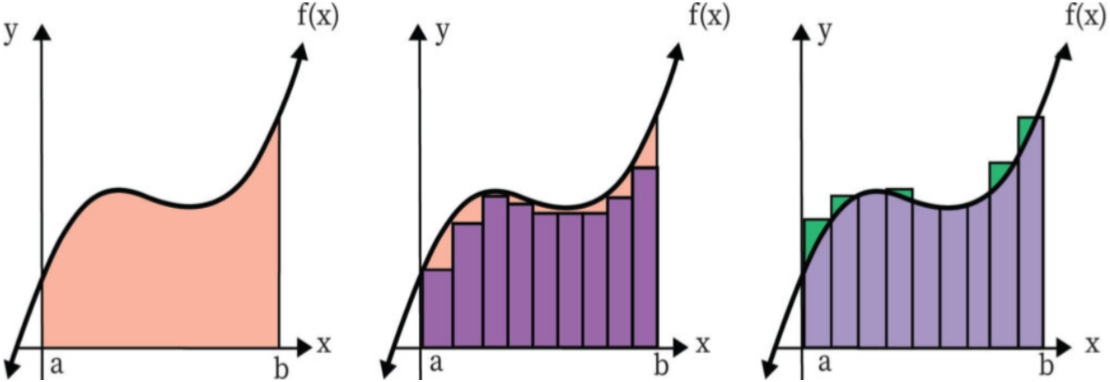
- Define the **lower** (Darboux) **sum** $L(P, f)$ and the **upper** (Darboux) **sum** $U(P, f)$ as:

$$L(P, f) = \sum_{i=1}^{n} m_i(x_i - x_{i-1}) \qquad U(P, f) = \sum_{i=1}^{n} M_i(x_i - x_{i-1})$$

Note that each term in the sum is the area of a rectangle.

The Riemann Integral

Upper and Lower Sums



Motivation

The Riemann Integral

- Define the **lower integral** $I_*(f)$ and **upper integral** $I^*(f)$ on $[a, b]$ as:

$$I_*(f) = \sup_P L(P, f) \qquad I^*(f) = \inf_P U(P, f)$$

Theorem 3.1 (Riemann Integrability)

Let f be a non-negative, real-valued, and bounded function on the interval $[a, b]$. f is Riemann Integrable if and only if

$$I(f) = \int_a^b f(x)\, dx = I_*(f) = I^*(f)$$

i.e., if the upper and lower integrals are the same for any choice of partition P. $I(f)$ is the area under f in the interval $[a, b]$.

Nice illustration. We can now address the initial problem.

Motivation

Limitations of the Riemann Integral

- Usually the definition is difficult to apply (any P!), so we use the **Fundamental Theorem of Calculus** and compute antiderivatives. This makes integration way easier!
- Remember our problem: $\mathbb{P}(X \in \mathbb{Q}) = \int_0^1 \mathbb{1}_{\mathbb{Q}}(x)\ dx$. Can we use the Riemann integral?

Answer (The Dirichlet Function is not Riemann Integrable)

$\int_0^1 \mathbb{1}_{\mathbb{Q}}(x)\ dx = \nexists$

Proof. *(Hint: Apply the definition of Riemann Integrability by taking an arbitrary partition P using the fact that $\mathbb{Q}$ is dense in $\mathbb{R}$).*

- Can we still compute the probability? Yes, but we need another integral: the **Lebesgue Integral**.
- For this task, we need to introduce more mathematical machinery.

Relation to Measure Theory

Measurable Functions

- ***Definition 3.1***: Let $f : \Omega \to B$ be a function. Let $D \subseteq B$. The **inverse image** or **pre-image** of f as the set $f^{-1}(D) := \{\omega \in \Omega : f(\omega) \in D\}$. *Example.*
- **Claim 3.1:** Let $D_1, \ldots, D_n$ be subsets of B. Then:
  1. $f^{-1}(D^c) = \left(f^{-1}(D)\right)^c$
  2. $f^{-1}(\cup_n D_n) = \cup_n f^{-1}(D_n)$

  Proof.
- ***Definition 3.2***: Let $(\Omega, \mathcal{F})$ and $(\Lambda, \mathcal{G})$ be measurable spaces and let $f(\omega)$ be a function $f : \Omega \to \Lambda$. We say that f is a **measurable function** from $(\Omega, \mathcal{F})$ to $(\Lambda, \mathcal{G})$ if

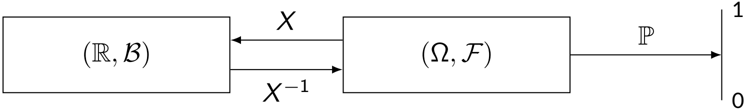
$$f^{-1}(G) = \{\omega \in \Omega : f(\omega) \in G\} \in \mathcal{F} \ , \ \forall G \in \mathcal{G}$$

Examples.

Relation to Measure Theory

Measurable Functions

- If $\Lambda = \mathbb{R}$ and $\mathcal{G} = \mathcal{B}$, $f(\omega)$ is said to be **Borel-measurable** (Borel function).
- **Claim 3.2:** Let f be measurable from from $(\Omega, \mathcal{F})$ to $(\Lambda, \mathcal{G})$. Then $f^{-1}(\mathcal{G}) := \{f^{-1}(G) : G \in \mathcal{G}\}$ is a sub-σ-field of $\mathcal{F}$. ***Proof.***
- **Claim 3.3:** The function $f : \Omega \to \Lambda$ is measurable from $(\Omega, \mathcal{P}(\Omega))$ to $(\Lambda, \mathcal{G})$. ***Proof.***
- If $X(\omega)$ is measurable from $(\Omega, \mathcal{F})$ to $(\mathbb{R}, \mathcal{B})$ then $X(\omega)$ is a **random variable**.



- ***Exercise*:** Prove that the rational numbers are in the Borel σ-algebra, i.e., $\mathbb{Q} \in \mathcal{B}$. Use the definition of $\mathcal{B}$ from Day 1 and use the fact that $\mathbb{Q}$ is countable.
  Is the Dirichlet function measurable?

Relation to Measure Theory

Measures

- ***Definition 3.2*:** Let $(\Omega, \mathcal{F})$ be a measurable space. A set function $\mu : \mathcal{F} \to [0, \infty]$ is a **measure** if it satisfies

  1. $\mu(\varnothing) = 0$
  2. If $A_1, A_2 \dots \in \mathcal{F}$ and $A_i \cap A_j = \varnothing$ for $i \neq j$, then $\mu\left(\cup_n A_n\right) = \sum_n \mu(A_n)$.

- ***Example 3.1*:** Let $(\Omega, \mathcal{F})$ be a measurable space and $A \in \mathcal{F}$. The **Dirac measure** or point mass at c is given by

$$\delta_c(A) = \begin{cases} 0, & c \notin A \\ 1, & c \in A \end{cases}$$

- ***Example 3.2*:** Let $(\Omega, \mathcal{F})$ be a measurable space and let $\nu(A)$ be the number of elements in $A \in \mathcal{F}$ ($\nu(A) = \infty$ if A contains infinitely many elements). Then, ν is the **counting measure** on $\mathcal{F}$. It is also the countable sum of Dirac measures: $\sum_{c \in A} \delta_c(A)$.

- ***Example 3.3*:** There is a unique measure λ on $(\mathbb{R}, \mathcal{B})$ that satisfies $\lambda([a, b]) = b - a$ for any finite interval $-\infty < a \leqslant b < \infty$. This measure λ is the **Lebesgue measure**.
- ***Definition 3.6*:** If a measure $\mathbb{P}$ satisfes $\mathbb{P}(\Omega) = 1$, it is called **probability measure**. Equivalently, if we restrict λ to the measurable space $([0, 1], \mathcal{B}_{[0,1]})$, the resulting measure $\mathbb{P}$ is the probability measure.
- The triplet $(\Omega, \mathcal{F}, \mu)$ is a **measure space**. Does the definition of measure ring a bell? Recall the Axioms on $\mathbb{P}$.

Relation to Measure Theory

Integration

- ***Definition 3.3***: A **simple function** $\phi : \Omega \to \mathbb{R}$ is a linear combination of indicators $\mathbb{1}_A(\omega)$ of measurable sets $A_1, A_2, \ldots \in \mathcal{F}$, i.e.,

$$\phi(\omega) = \sum_{i=1}^{n} a_i \mathbb{1}_{A_i}(\omega)$$

  where $a_1, a_2, \ldots \in \mathbb{R}$ and $A_1, A_2, \ldots \subseteq \Omega$.

- ***Definition 3.6***: The **positive part of a function** f is given by $f^+(\omega) := \max\{f(\omega), 0\}$. The **negative part of a function** f is given by $f^-(\omega) := \max\{-f(\omega), 0\}$.

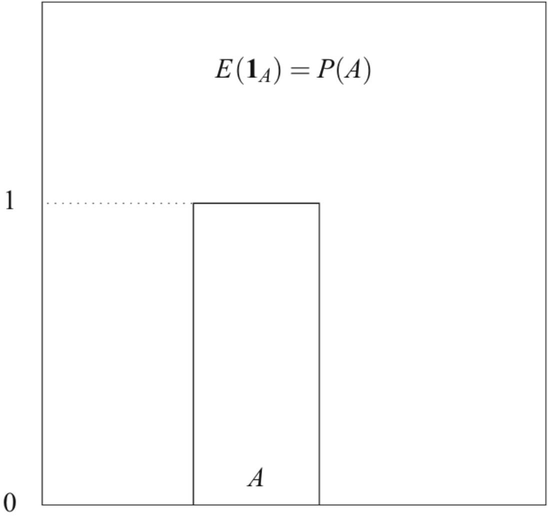
Theorem 3.2 (Pointwise Limit of Simple Functions)

Let $(\Omega, \mathcal{F})$ be a measurable space and let $f : \Omega \to \mathbb{R}_+$ be a positive $\mathcal{F}$-measurable function. Then, there exists a non-decreasing sequence $\{\phi_n\}_{n \in \mathbb{N}}$ of simple functions such that $\forall \omega \in \Omega$ $\lim_{n\to\infty} \phi_n(\omega) = f(\omega)$.

Relation to Measure Theory

Integration

We now introduce the construction of the **abstract integral**, i.e., w.r.t. any measure μ.



***Definition 3.7*:** (Four Steps)

- Step 1: Indicator Functions

$$\mathbb{1}_A = \mathbb{1}_{\{\omega \in A\}}$$

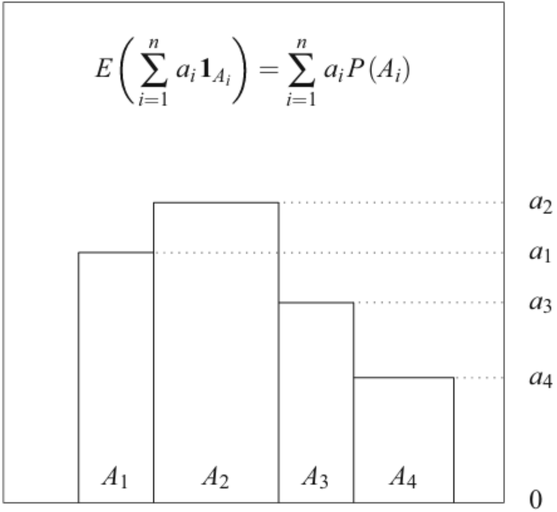
Think about integrals as measures of functions. It is natural to think about the measure of the indicator as the area of the rectangle: *height* $\times$ *length*. The height is 1, but A can be very abstract, so we use its measure instead. Hence:

$$\int_\Omega \mathbb{1}_A d\mu = \mu(A)$$

Relation to Measure Theory

Integration

We now complicate a bit more the function to be integrated.



– Step 2: Simple Functions $\phi : \Omega \to \mathbb{R}$ with finite range.

$$\phi(\omega) = \sum_i a_i \mathbb{1}_{A_i}(\omega)$$

They are just linear combinations of indicators. We can then add up the areas of the different rectangles. Hence:

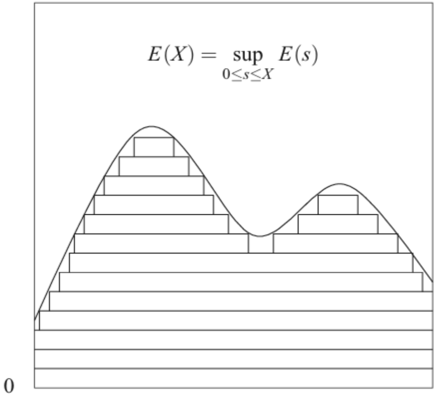
$$\int_\Omega \phi(\omega)\, d\mu = \int_\Omega \sum_i a_i\, \mathbb{1}_{A_i}\, d\mu$$

$$= \sum_i a_i \int_\Omega \mathbb{1}_{A_i}\, d\mu = \sum_i a_i\, \mu(A_i)$$

Relation to Measure Theory

Integration

And another layer of complication...



- <u>Step 3: Positive Measurable Functions</u>

$$X(\omega) \geqslant 0$$

Let $\mathcal{S}_X$ be the collection of all non-negative simple functions such that $\phi(\omega) \leqslant X(\omega)\ \forall \omega \in \Omega$. The integral of X with respect to μ is defined as:

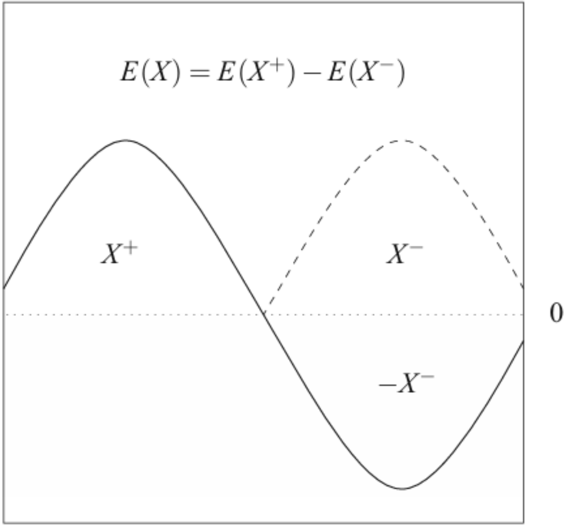
$$\int_{\Omega} X \, d\mu = \sup \left\{ \int \phi \, d\mu : \phi \in \mathcal{S}_X \right\}$$

Hence, for all X there exists a sequence of simple functions $\phi_1, \phi_2, \ldots$ such that $0 \leqslant \phi_i \leqslant X\ \forall i$ and $\lim_{n\to\infty} \int \phi_n \, d\mu = \int X \, d\mu$

Relation to Measure Theory

Integration

And finally:



- Step 4: Integrable Functions They are much more general than Riemann integrable functions. This one is easy, we just integrate the positive and negative parts separately:

$$\int_\Omega X \, d\mu = \int_\Omega X^+ \, d\mu - \int_\Omega X^- \, d\mu$$

Problem: if both integrals are ∞, we have an indetermination ($\infty \cdot 0 = 0$ by convention). Thus, we need:

$$\int_\Omega |X| \, d\mu < \infty \quad \text{(Integrability)} \qquad \square$$

Relation to Measure Theory

Integration

- We can now define the **Lebesgue Integral**, i.e., the integral w.r.t. the Lebsegue Measure, λ :

$$\int_{\Omega} X \, d\lambda = \int_{\Omega} X(\omega) \, d\lambda(\omega) = \int_{\Omega} X(\omega) \, \lambda(d\omega)$$

  All functions that are Riemann integrable are also Lebesgue integrable, in which case the Lebesgue integral is equal to the Riemann integral.

- Initial problem: $\mathbb{P}(X \in \mathbb{Q}) = \int_0^1 \mathbb{1}_{\mathbb{Q}}(x) \, dx = \nexists$. We have shown that the Dirichlet function is measurable. We could thus integrate it w.r.t. the Lebesgue measure.

Answer (Lebesgue Integral of the Dirichlet Function)

$\int_0^1 \mathbb{1}_{\mathbb{Q}}(x) \, d\lambda = 0$

Proof. *(Hint: Use Integration Step 1, the definition of Lebesgue measure, and the fact that $\mathbb{Q}$ is countable so that it can be expressed as a countable union of disjoint singletons).*

Relation to Measure Theory

Examples of Integration under some Important Measures

- **Claim 3.4** Let $(\Omega, \mathcal{F}, \delta_c)$ be a measure space and let $X : \Omega \to \mathbb{R}$ be a function. Then

  $$\int_{\Omega} X(\omega)\ d\delta_c(\omega) = X(c)$$

  Proof.

- **Claim 3.5** Let $(\Omega, \mathcal{F}, \nu)$ be a measure space where Ω is countable and $\mathcal{F} = \mathcal{P}(\Omega)$. Let $X : \Omega \to \mathbb{R}$ be a function. Then

  $$\int_{\Omega} X(\omega)\ d\nu(\omega) = \sum_{\omega \in \Omega} X(\omega)$$

  Proof.

- **Claim 3.6** If $\Omega = \mathbb{R}$ and $\mathcal{F} = \mathcal{B}$, then the integral of $X(\omega)$ w.r.t. λ over an interval $[a, b]$ agrees with the Riemann integral in calculus when the latter is well defined:

  $$\int_{[a,b]} X(\omega)\ d\lambda(\omega) = \int_a^b X(\omega)\ d\omega$$

Relation to Measure Theory

Integration

We can now translate all this into the context of probability.

- Measurable Sets, $A \equiv$ Events.
- Measurable Functions, $X(\omega) \equiv$ Random Variables. $X(\omega) : (\Omega, \mathcal{F}, \mathbb{P}) \underset{\omega \mapsto x}{\longrightarrow} (\mathbb{R}, \mathcal{B})$
- General Measure, $\mu \equiv$ Probability Measure, $\mathbb{P}$.
- Generic Measure Space, $(\Omega, \mathcal{F}, \mu) \equiv$ Probability Space $(\Omega, \mathcal{F}, \mathbb{P})$.
- ***Definition 3.8***: The **Expectation** of a random variable X is the integral with respect to the Probability Measure, $\mathbb{P}$, in the measurable space $(\Omega, \mathcal{F})$ i.e.,

$$\mathbb{E}[X] = \int_{\Omega} X \, d\mathbb{P}$$

Problem: How do we compute the integral in Definition 3.8? It seems difficult to handle $\mathbb{P}$, right? It is. We need another important and abstract tool (yes, another one).

The rest of the story. If time permits...

Relation to Measure Theory

The Radon-Nikodým Derivative and the Change of Variables Formula

- To simplify the computation of an integral w.r.t. the probability measure we need the notion of **Distribution** of a random variable. Recall that a random variable is a measurable function:

  $$X(\omega) : (\Omega, \mathcal{F}, \mathbb{P}) \underset{\omega \mapsto x}{\longrightarrow} (\mathbb{R}, \mathcal{B})$$

  And the (mysterious) probability measure $\mathbb{P}$ is defined in $(\Omega, \mathcal{F})$. It would be easier to integrate in $\mathbb{R}$ by looking at "events" $B \in \mathcal{B}$ and applying typical integration methods (Fundamental Theorem of Calculus). Do we have an equivalent measure we can use for that purpose that yields the same results?

- For this we need two things: the **Radon-Nikodým Theorem** and the **Change of Variables Formula**. These two tools will allow us to introduce the definition of distributions and to safely switch spaces and measures when computing expectations.

- But first we need some (abstract) definitions.

Relation to Measure Theory

The Radon-Nikodým Theorem: Setup

- ***Definition 3.9***: A measurable measure space is said to be σ-**finite** whenever Ω can be written as a countable union of measurable sets with finite measure. *(Any finite μ).*

- ***Definition 3.10***: A measure ν is said to be **Absolutely Continuous** with respect to another measure μ, written $\nu \ll \mu$, whenever

$$\forall\ A \in \mathcal{F},\ \mu(A) = 0 \implies \nu(A) = 0. \quad \textit{Examples.}$$

- To motivate the theorem, let μ be a measure and let $\phi \geqslant 0$ be a measurable function from $(\Omega, \mathcal{F}, \mathbb{P})$ to $(\mathbb{R}, \mathcal{B})$. Now define a new function on $\mathcal{F}$:

$$\nu(A) = \int_A \phi\, d\mu = \int_\Omega \phi \mathbb{1}_A\, d\mu \quad \forall\ A \in \mathcal{F}$$

  This is actually a new measure, e.g., if $\phi(\omega) = 1$ then $\nu(A) = \mu(A)$. So you can view ϕ as a function that will "deform" the measure μ and will end up being another measure.

Relation to Measure Theory

The Radon-Nikodým Theorem: Setup

- **Exercise: Prove that $\nu(A)$ above is a measure.** *(Hint: Use the definition of measure).*
- We thus know that given μ, ϕ, and A, we can integrate and get a new measure ν.
- The **Radon-Nikodým Theorem** tries to answer: given μ, A, and ν, can we find $\phi \geqslant 0$ that satisfies:

  $$\nu(A) = \int_A \phi \, d\mu$$

  Which is way more difficult! And the theorem gives both existence and uniqueness!
- The RN Theorem says that an absolute continuity condition is **necessary and sufficient** for the existence and uniqueness of $\phi \geqslant 0$.

Relation to Measure Theory

The Radon-Nikodým Theorem

Theorem 3.2 (Radon-Nikodým Theorem)

Let μ and ν be two σ-finite measures such that $\nu \ll \mu$. Then,

1. There exists a positive measurable function $\phi : (\Omega, \mathcal{F}) \longrightarrow (\mathbb{R}, \mathcal{B})$ such that:

$$\nu(A) = \int_A \phi \, d\mu \ \ \forall A \in \mathcal{F}$$

   The function ϕ is written $\phi = d\nu/d\mu$ and is called the Radon-Nikodým derivative of measure ν with respect to μ.
2. The function ϕ is unique almost everywhere μ (a.e.-μ) (i.e., all the sets in which they differ have measure μ zero).

We will not prove the theorem.

- How does this help with expectations? We can now define the concept of distribution.

Relation to Measure Theory

Induced Measure and Distribution

- Recall that $X : (\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow (\mathbb{R}, \mathcal{B})$. Even though $\mathbb{P}$ is too complex, we can define a probability measure $\nu_X(B)$ on $\mathcal{B}$.
- ***Definition 3.11*:** For a random variable X, define the measure

$$\nu_X(B) = \mathbb{P}(X \in B) = \int_\Omega \mathbb{1}_{\{\omega \in X^{-1}(B)\}} \, d\mathbb{P} \quad \forall B \in \mathcal{B}$$

  This is called **Induced Measure** in measure theory and **Distribution** in probability theory. $\nu_X(B) : (\mathbb{R}, \mathcal{B}) \underset{x \mapsto p}{\longrightarrow} [0, 1]$.
- If we evaluate the distribution $\nu_X(B)$ at the Borel set $B = (-\infty, x]$, we get the **CDF** we saw before (check this). This definition is thus much more general and abstract.
- Instead of looking at probabilities $\mathbb{P}$ of events, we look at Borel sets $B \in \mathcal{B}$ and then look X^{-1}, which takes us back to $\mathcal{F}$. We can thus use it for integration: Change of Variables Formula.

Relation to Measure Theory

The Change of Variables Formula

Theorem 3.3 (Change of Variables Formula)

Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow (\mathbb{R}, \mathcal{B})$ and let $h : (\mathbb{R}, \mathcal{B}) \longrightarrow (\mathbb{R}, \mathcal{B})$ be measurable and integrable. Then,

$$\mathbb{E}[h(X)] = \int_{\mathbb{R}} h \, d\nu_X = \int_{\mathbb{R}} h(x) \, d\nu_X(x)$$

Proof.

- We can compute probabilities of real-valued events, i.e., $\mathbb{P}(X \in B)$, and the expectation of a random variable, i.e., $\mathbb{E}[X]$, just by taking $h(X) = \mathbb{1}_B$ and $h(X) = X$.
- We can now construct PDFs and PMFs. We can do this by checking an absolute continuity condition and applying the Radon-Nykodým Theorem. This condition will be slightly different for continuous and discrete random variables.

Relation to Measure Theory

PDFs and PMFs

- A random variable X is said to be **continuously distributed** whenever its distribution ν_X is **absolutely continuous with respect to the Lebesgue measure** λ. Since λ is σ-finite, the RN Theorem applies and gives the existence and uniqueness of a measurable function $\phi(x)$ such that $d\nu_x = \phi \, d\lambda$. This function is the **PDF** of X. Hence:

$$\mathbb{E}[h(X)] = \int_{\mathbb{R}} h \, d\nu_X = \int_{\mathbb{R}} h \, \phi \, d\lambda = \int_{\mathbb{R}} h(x)\phi(x) \, dx$$

- A random variable is said to be **discrete** whenever its range S is either finite or countable, in which case its distribution is **absolutely continuous with respect to the counting measure** on S, i.e., $\nu_X \ll \mu_A$ where $\mu_A = card(A)$ for all $A \subseteq S$. By the RN Theorem, there exists a unique non-negative measurable function $\phi(x)$ such that $d\nu_X = \phi \, d\mu_S$. This function is the **PMF** of X. Hence:

$$\mathbb{E}[h(X)] = \int_{\mathbb{R}} h \, d\nu_X = \int_{\mathbb{R}} h \, \phi \, d\mu_S = \sum_{x \in S} h(x)\phi(x)$$

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